OENG1209: Engineering Science

Week 3:

Newton's Laws of Motion

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Today's Topics

- **Forces:** definitions, characteristics, unit.
- **Newton's Laws of Motion:** how to quantify forces and the laws that govern motion.
- **Free Body Diagram:** the road map to your solution.
- **Types of Forces:** how to model different physical forces.

Forces

Definition and Representations

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Forces– Definition

- **A force** is a push or a pull. There are many types of forces but here we will think of a force as a very specific action.





Forces– Definition

- A force **must act on an object**. In other words, the pushes and pulls are applied to something. From the object's perspective, it has a force **exerted** on it.





Forces– Definition

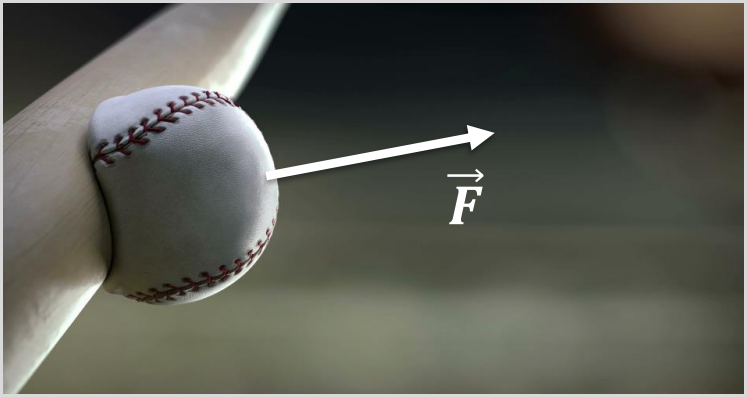
- All forces require **an agent**, something that acts or exert power. If a force is being exerted on an object, you must be able to identify a specific cause.





Forces– Definition

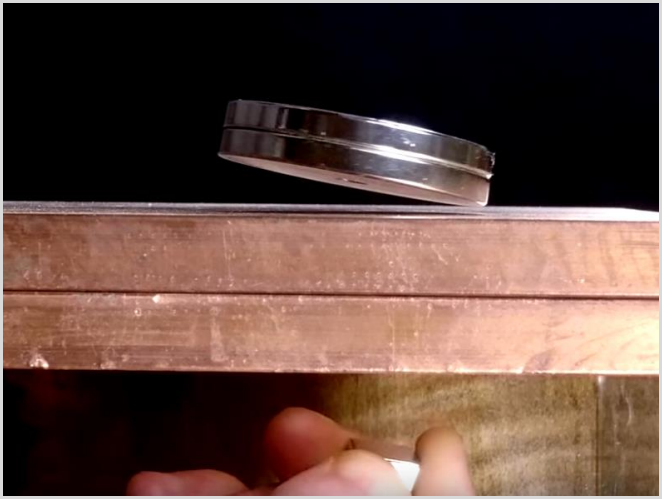
- **A force** is a vector. This means all forces have magnitudes and directions.



Forces– Definition

- A force can be either a **contact force** (exerted via direct contact) or a **long-range force** (exerted without contact).

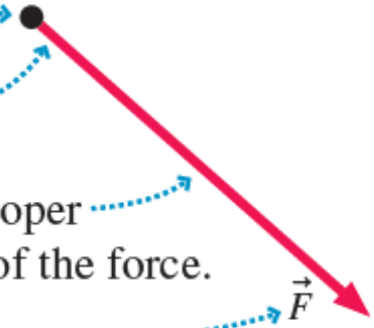




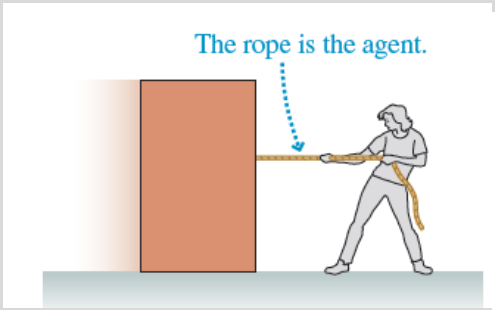
Forces– Representation

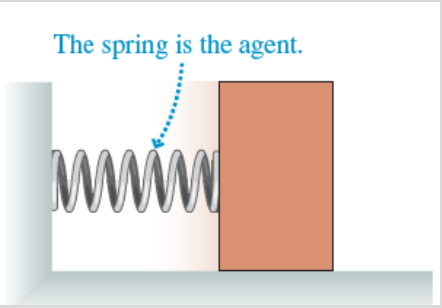
Drawing force vectors

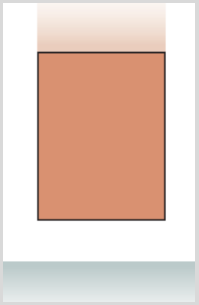
1. Model the object as a particle.
2. Place the *tail* of the force vector on the particle.
3. Draw the force vector as an arrow pointing in the proper direction and with a length proportional to the size of the force.
4. Give the vector an appropriate label.

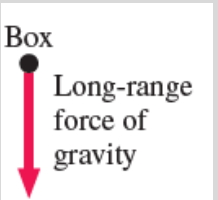


Forces– Representation Examples

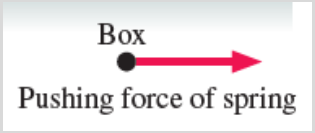










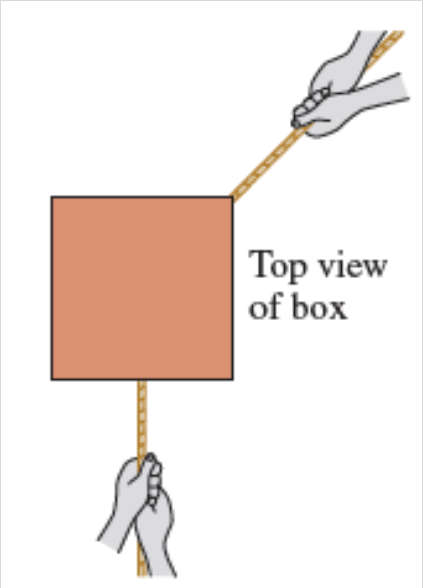


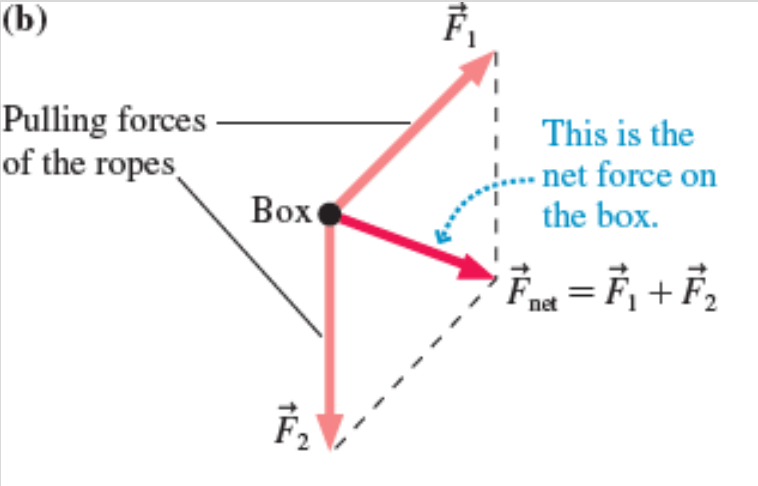
Forces– Combining Forces

- Forces are combined via **Vector addition**.
- If we have several forces $\overrightarrow{F_1}, \overrightarrow{F_2}, \overrightarrow{F_3}$, ... exerting on the same object, the net force resulting from those forces will be:

$$\overrightarrow{F_{net}} = \overrightarrow{F_1} + \overrightarrow{F_2} + \overrightarrow{F_3} + \cdots + \overrightarrow{F_n}$$

Forces– Combining Forces Example

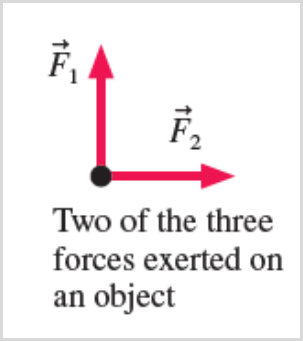




Parallelogram rules to add the force

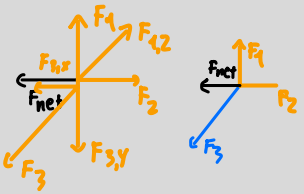
Forces– Exercise

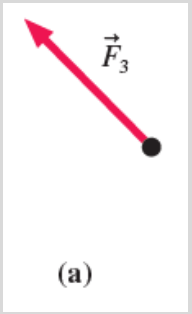
- Two of the three forces exerted on an object are shown. The net force points to the negative $x - direction$. Which is the missing third force?

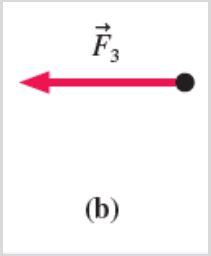


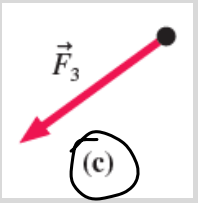
Two of the three forces exerted on an object

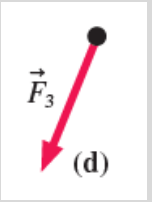
$F_{net} = 0 = F_1 + F_2 + F_3$





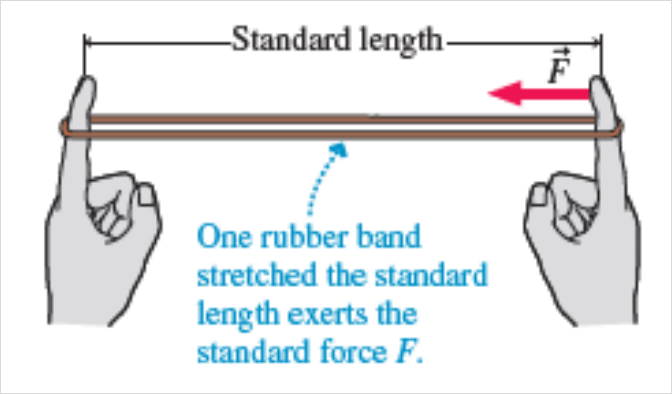






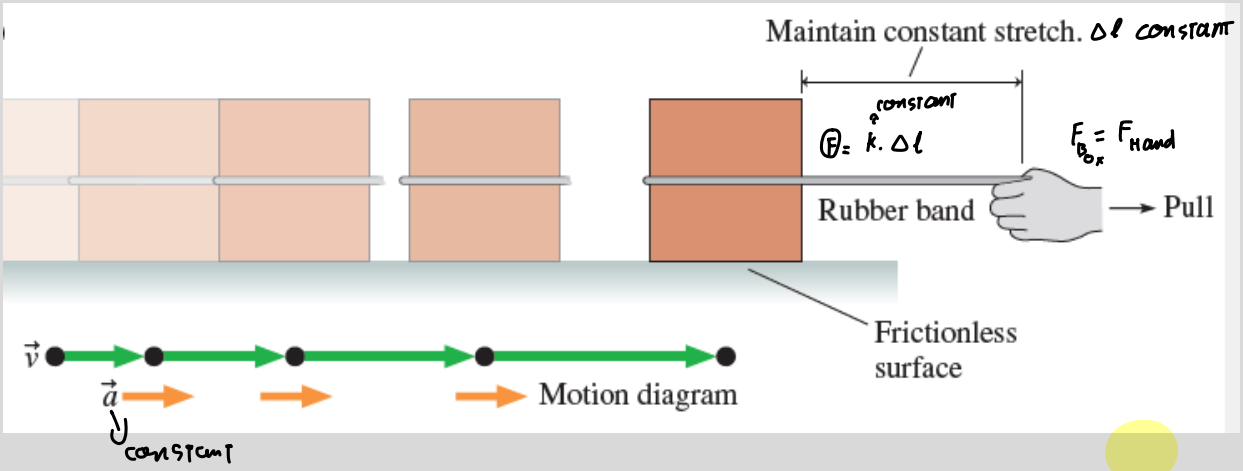
Forces– What does Forces Do?

- Consider a rubber band being pulled apart between your fingers.



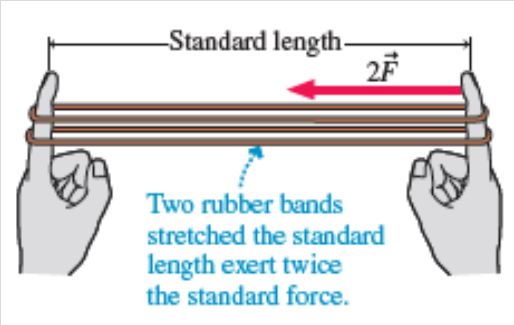
Forces– What does Forces Do?

- Now consider the case where the rubber band is stretched and attached to a block of weight resting on a frictionless surface:



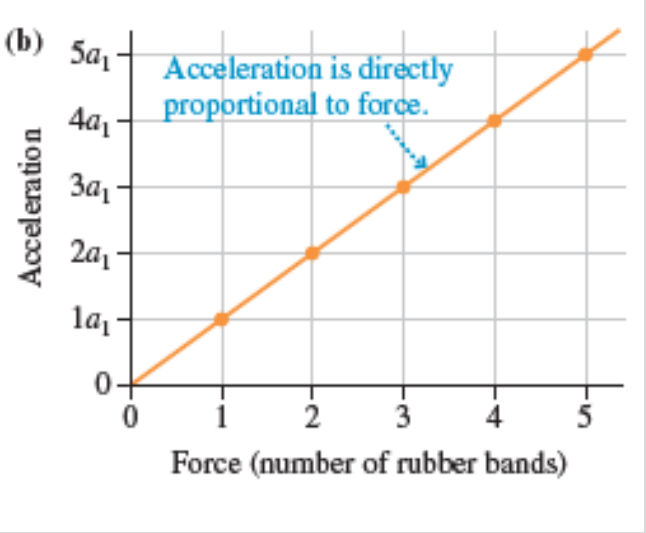
Forces– What does Forces Do?

- **Conclusion #1:** an object pulled with a constant force moves with a constant acceleration.
- A force causes an object to **accelerate** and a constant force produces a **constant acceleration**.
- What if you increase the number of rubber bands?



Forces– What does Forces Do?

- We can easily observe that the amount of force exerted on the block is **directly proportional** to the number of rubber bands. Recall our Conclusion #1, we can plot the acceleration vs. force relationship as:



$a \propto F$

$\uparrow F = m a \uparrow$

$\downarrow$ constant

Forces– What does Forces Do?

- **Conclusion #2:** Acceleration is **directly proportional to the force**.

  Mathematically, this can be expressed as:

$$|\vec{a}| = c\left|\vec{F}\right|$$

  Where c is the slope of our graph.

Forces– Unit of Force

- In standard SI units, the unit of Force is **Newton (N)** with the following conversion:

$$\mathbf{1\ N = 1\ \frac{kg.m}{s^2}}$$

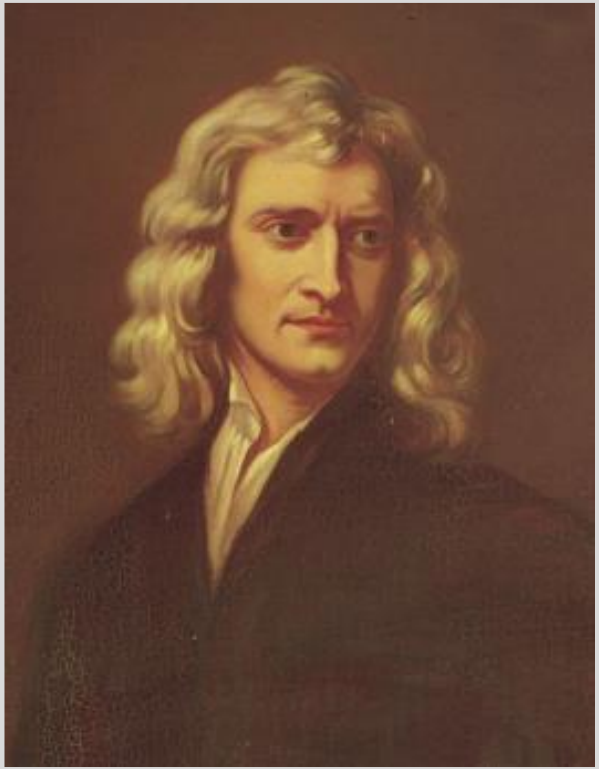
- In other words, $1\ N$ is the amount of Force needed to accelerate an object of $1\ kg - mass$ by $1\ m/s^2$.

Newton's 1st and 2nd Law of Dynamics

From falling apple to rocket science

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Sir Isaac Newton



Inertia of rest hard to move while rest
Direction → go straight
hard to change to left or right
motion
↓
hard to stop move

Newton's First Law

- **Newton's 1st Law:** An object that is at rest will remain at rest, or an object that is moving will continue to move in a straight line with **constant velocity**, if and only if the **net force** acting on the object is zero.

$$\overrightarrow{v_{object}} = constant,$$ either 0 or non 0

in motion or at rest

$$if \sum(\vec{F}) = 0$$

Newton's First Law – Mechanical Equilibrium

- **Mechanical equilibrium** is defined as the state of an object, during which its net force is zero. Objects in mechanical equilibrium have no acceleration, i.e. $\vec{a} = 0$
- Mathematical definition of mechanical equilibrium:

$$\sum(\vec{F}) = 0$$

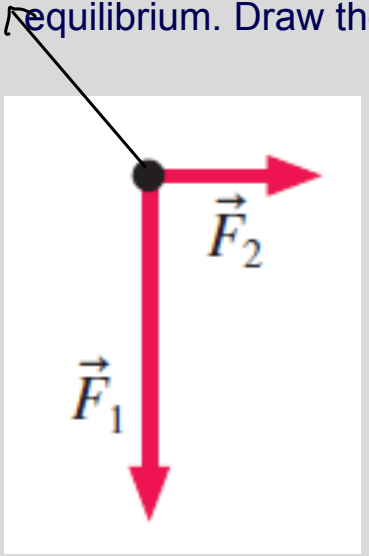
$$\overrightarrow{F_x} = 0 \; and \; \overrightarrow{F_y} = 0 \; and \; \overrightarrow{F_z} = 0$$

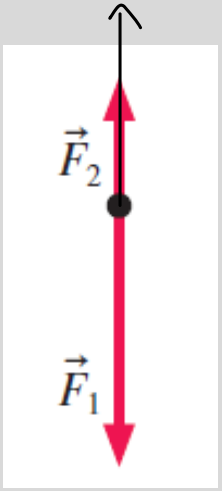
Newton's First Law – Purpose

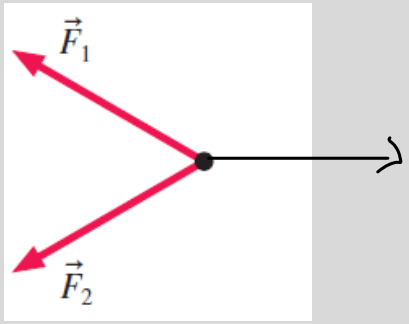
- What causes an object to change its velocity?
  - A force is what causes an object to change its velocity.
- The purpose of Newton's First Law is to identify **a force** as something that disturbs a state of equilibrium.

Newton's First Law – Example

- **Example:** The diagrams below showing 2 out of 3 forces acting on an object in equilibrium. Draw the vector of the 3rd Force and label it $\overrightarrow{F_3}$.







Mass and Inertia

- Before we go into **Newton's Second Law**, it is important to define our understanding of what is **Mass** and **Inertia**.

- **Mass** is the amount of matter an object contains. The more matter an object has, the more it resists accelerating in response to a force.





Mass and Inertia

quán tính

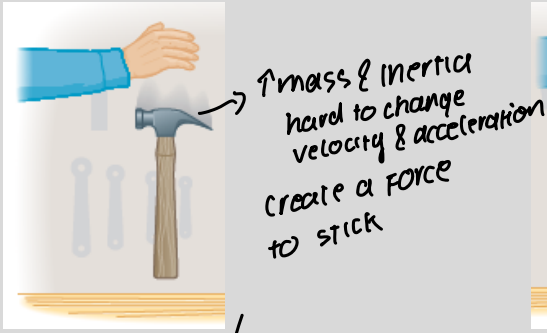
- **Inertia** is the tendency of an object to resist a change in its velocity (i.e. to resist acceleration).
- The higher the mass of an object is, the higher its inertia is.
- Unlike forces, mass and inertia are **scalar quantities**.



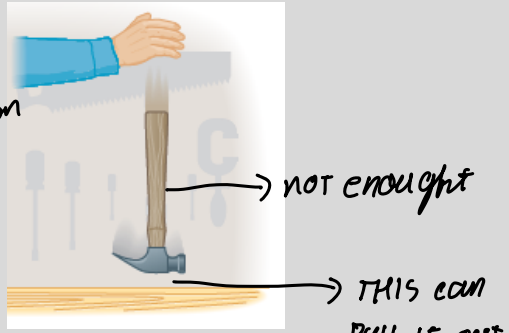


Mass and Inertia – Example

- **Example:** The metal head of a hammer is loose. To tighten it, you want to drop the hammer down onto a rigid table surface. Should you:
  a) Drop the hammer with the hammer end down?
  b) Drop the hammer with the head end down?
  c) Both options are the same.



Option (a)



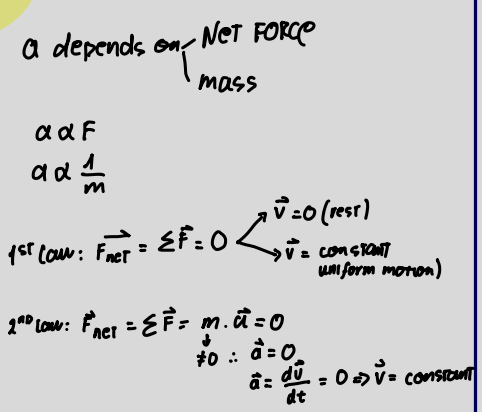
Option (b)

Newton's Second Law

- An object of **mass m** subjected to forces $\overrightarrow{F_1}, \overrightarrow{F_2}$, ... will undergo an acceleration $\vec{a}$ given by:

$$\vec{a} = \frac{\overrightarrow{F_{net}}}{m}$$

Where $\overrightarrow{F_{net}} = \overrightarrow{F_1} + \overrightarrow{F_2} + \overrightarrow{F_3} + \cdots$

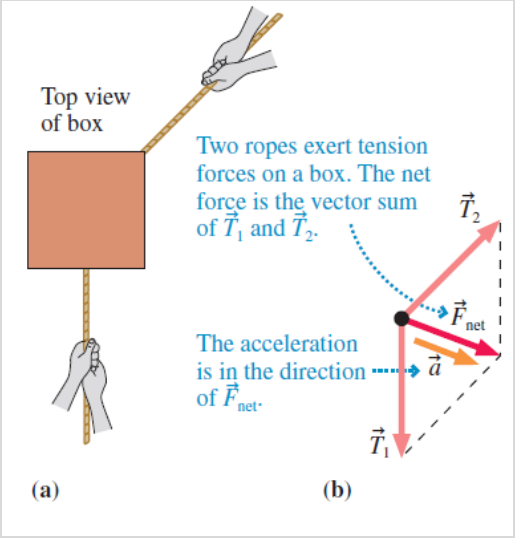


- The acceleration vector $\vec{a}$ points in the **same direction** as the net force vector $\overrightarrow{F_{net}}$.

Newton's Second Law – Purpose

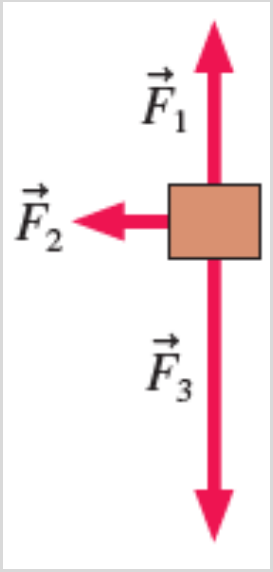
- **Newton's Second Law** formally establishes a **relationship between Force, Mass, and Acceleration**.

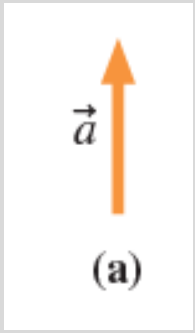
- The critical idea is that an object **accelerates in the direction of the net force vector $\overrightarrow{F_{net}}$.**

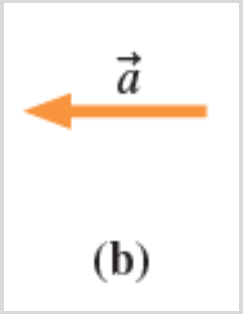


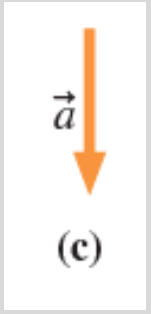
Newton's Second Law – Example 1

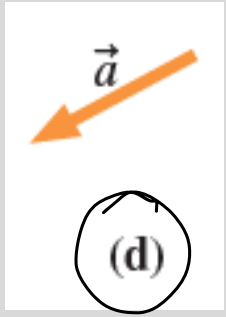
- **Example:** Three forces act on an object. In which direction does the object accelerate?

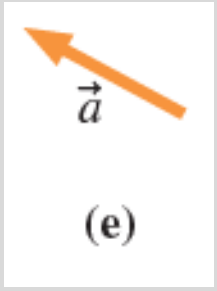






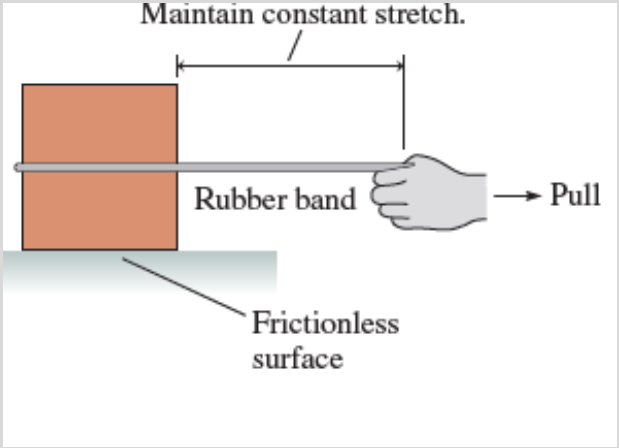






Newton's Second Law – Example 2

- **Example:** A constant horizontal force of 50N acts on a box on a smooth horizontal plane. If the box starts from rest and is observed to move 2.0 meters in 1.5 s, what is the mass of the body?



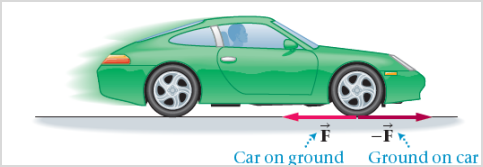
$m = F/a =$

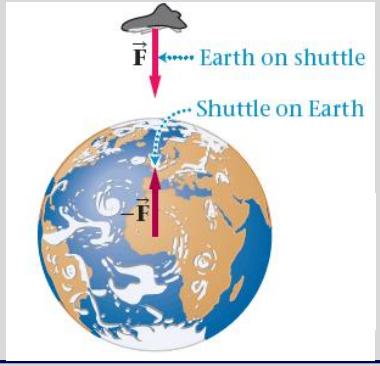
$\Delta x = v_0 t + \frac{1}{2} a t^2$

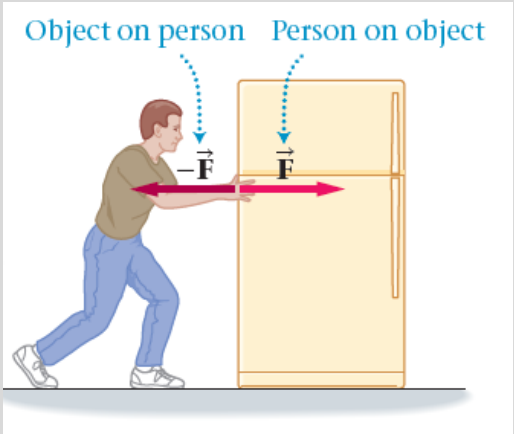
$2 = \frac{1}{2} a (1.5)^2$

Newton's Third Law

- **Newton's Third Law:** For every force that acts on an object, there is a **reaction force** acting on a different object that is equal in magnitude and opposite in direction







Newton's Third Law

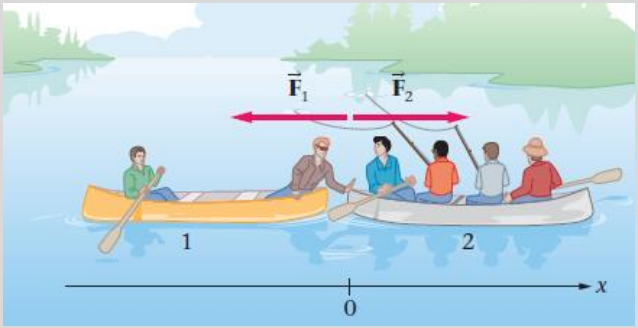
- Every force occurs as one member of an **action/reaction pair** of forces.
  - The two member of an action/reaction pair act on two different objects.
  - The two members of an action/reaction pair are equal in magnitude but opposite in direction: $\overrightarrow{F_{A\ on\ B}} = -\overrightarrow{F_{B\ on\ A}}$

Newton's Third Law – Purpose

- Newton's Third Law is important for solving Interacting – Object Problems.
- **Important:** even though the Action Force and Reaction Force are equal, the acceleration of each object may not be equal (due to Newton's 2nd Law).

Newton's 3rd Law – Example

- **Example:** Two groups of friends meet in the middle of a lake. After a brief chat, a person in canoe 1 pushes on canoe 2 with a force of 46 N to separate the canoes. Canoe 1 is carrying 2 people with total mass of 150 kg (including the canoe), and the total mass of canoe 2 with 4 people is 250 kg.
  a) What is the acceleration of each canoe after the push?
  b) What is the distance between the 2 canoes at 1.2 second after the push?



Free Body Diagram

Road Map to your solution

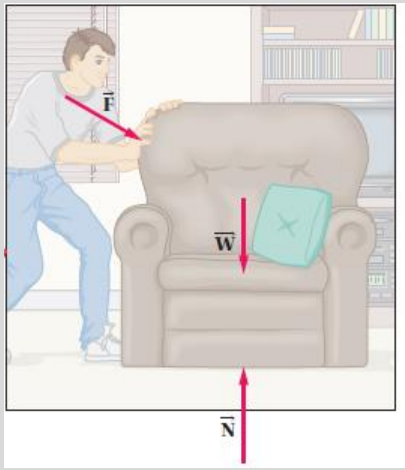
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Free Body Diagram

- **The Free Body Diagram** is a pictorial representation of a problem, represents the object as a particle and shows all of the forces acting on the object.

- Consider a man trying to move his sofa, can you identify the forces the sofa is feeling?



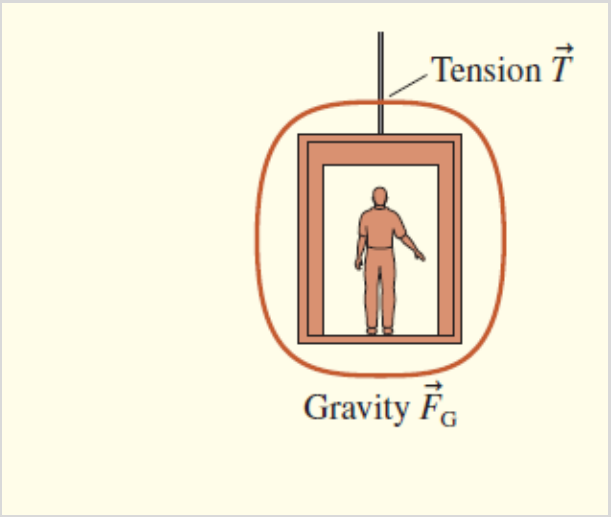


Free Body Diagram – How to draw

1. Identify all **forces** acting on the object.
2. Draw a **coordinate system**.
3. Represent the object as **a dot** at the origin of the coordinate axes.
4. Draw and label the Vectors representing each of **the identified forces**.
5. Draw and label the **net force vector** $\overrightarrow{F_{net}}$ besides the diagram. If $\overrightarrow{F_{net}} = \mathbf{0}$, then indicate so.

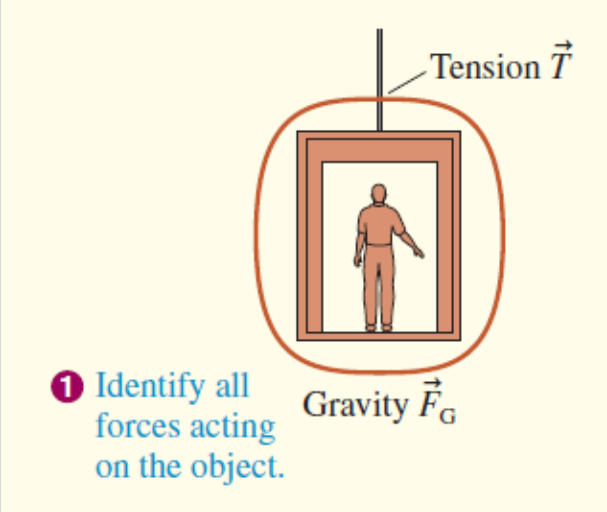
Free Body Diagram – Example

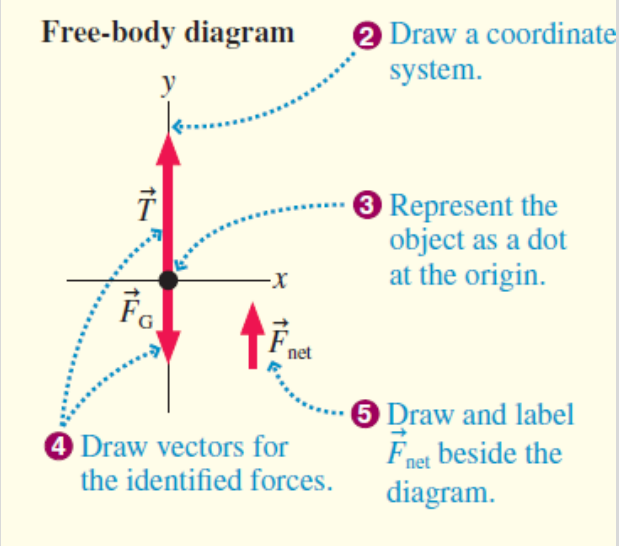
- **Example:** An elevator, suspended by a cable, speeds up as it moves upward from the ground floor. Identify the forces and draw a free-body diagram of the elevator.



Free Body Diagram – Example

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Problem Solving Strategy

1. **Draw a pictorial representation**.
   a) Model the object as a particle.
   b) Determine the object's acceleration vector $\vec{a}$.
   c) Identify all forces acting on the object and show them on a Free Body Diagram.
2. **Invoke Newton's 2nd Law:** $\overrightarrow{F_{net}} = m\vec{a}$. Then either:
   a) **Solve for the acceleration**, then use kinematics to find velocities and positions.
   b) **Use kinematics to determine the acceleration**, then solve for unknown forces.

Free Body Diagram – Example 2

- **Example:** A 500 grams model rocket with a weight of 4.90 N is launched straight up. The small rocket motor burns for 5.00 s and has a steady thrust of 20.0 N. What maximum altitude does the rocket reach?

$20 - 4.9 = 15.1\ N$, $\frac{15.1\ N}{0.5} = 30.2\ m/s^2$

$x = vt + \frac{1}{2}at^2 + x_0 = \frac{1}{2}at^2 = \frac{1}{2}(30.2)(5)^2 = 377.5\ m$

$4.9 \div 0.5 = 9.8\ m/s^2$

$v = at = 30.2(5) = 151\ m/s$

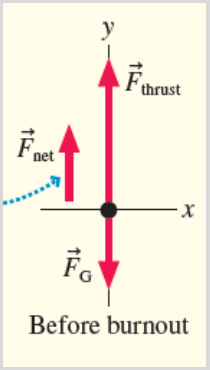
$v^2 = u^2 + 2a\Delta x$

$0^2 = (151)^2 + 2(-9.8)\Delta x$

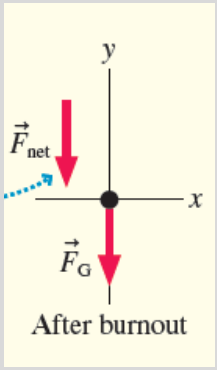
$\frac{-(151)^2}{2(9.8)} = 1163\ m$

$1163 + 377.5 = 1540.5$





Before burnout



After burnout

Types of Forces

How forces is generated and applied

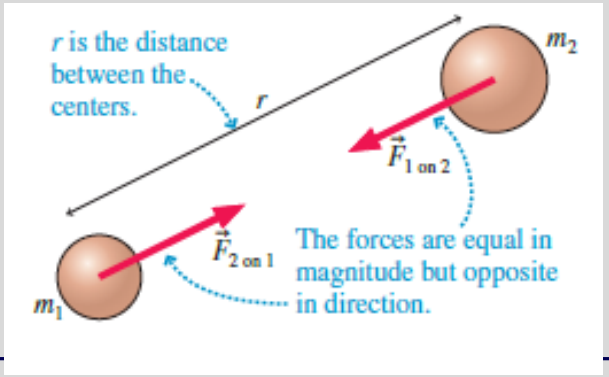
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Gravitational Forces

- Gravity is an **attractive**, **long-range force** between any two objects.
- Mathematically, the gravitational force between 2 objects with mass m_1 and m_2 is:

$$F_{1\ on\ 2} = F_{2\ on\ 1} = \frac{Gm_1m_2}{r^2}$$

Where G is the Gravitation Constant = $6.67 \times 10^{-11} Nm^2/kg^2$



Gravitational Forces

- According to the previous equation, we can calculate **the gravitational acceleration** of any planet by the equation:

$$g = \frac{GM}{R^2}$$

g = the gravitation acceleration on the planet, in m/s^2

G = the Gravitational Constant = $6.67 \times 10^{-11} Nm^2/kg^2$

M = the planet's Mass, in kilograms.

R = the planet's Radius, in meters.

Gravitational Forces – Example

- **Example:** Find the gravitational acceleration of the planet Earth, given that the Earth has average radius of $6.37 \times 10^6\ m$ and mass of 5.98×10^{24} kg.



Weight

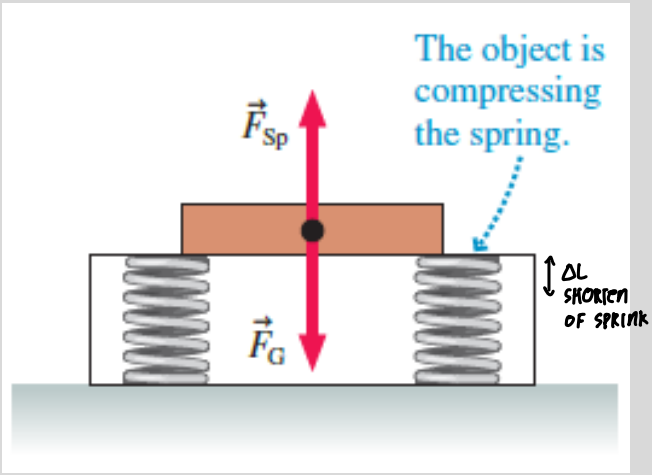
- **The Weight** of an object is a measure of how much force the object is exerting on the floor underneath it.
- By definition, **the Weight** of a stationary object is:

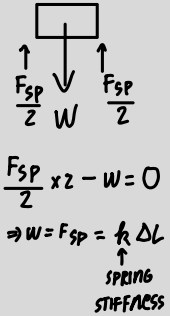
$$\boldsymbol{w = mg}$$

- Note that the Weight is defined by the gravitational acceleration g. Thus a person will "weight" differently on the Moon compared to on Earth.
- **Mass** and **Weight** are not the same thing. **Mass** is an intrinsic property of an object, while **Weight** is dependent on gravitational acceleration.

Weight

- The weight of an object is also often measured by a spring scale.
- Note that the scale does not "know" the weight of an object. What it can do is to measure **how much its spring is compressed** $(\overrightarrow{F_{SP}})$.







Weight vs. Spring

- The force $\overrightarrow{F_{SP}}$ exerted by the Spring on the object on top of the Scale is given by **Hooke's Law**:

$$\overrightarrow{F_{SP}} = -k\vec{\Delta s}$$

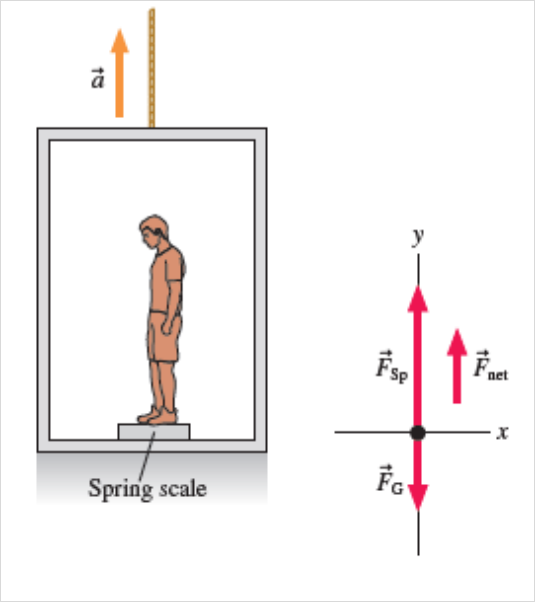
$\boldsymbol{k}$ = the Spring Constant specific to the Spring (N/m)

$\vec{\Delta s}$ = is the displacement vector of the Spring (m)

- Note that the direction of the Spring Force is **always opposite** the direction of the Displacement Vector.

Weight – Example

- **Example:** A man standing on a scale inside an elevator being pulled upward with an acceleration $\overrightarrow{a_y}$. Find an expression for the reading on the scale in terms of the acceleration of the elevator.



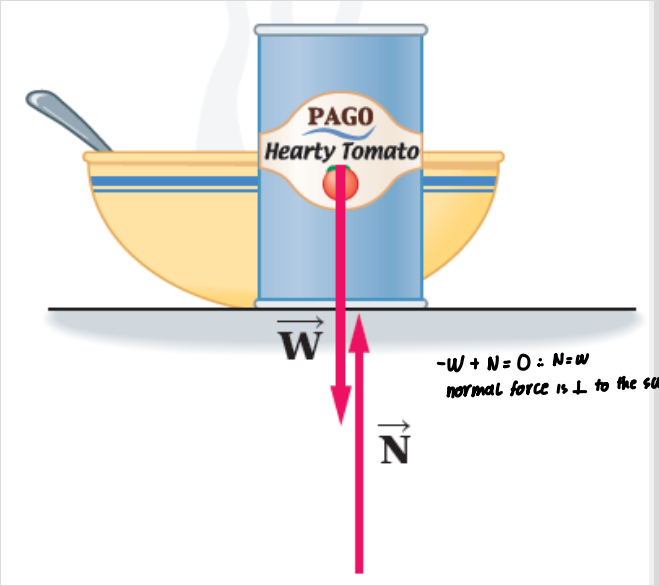
$F_{net} = F_{sp} + F_g = ma_y$

$F_{sp} = mg + ma_y = m(g + g_y)$

Normal Force

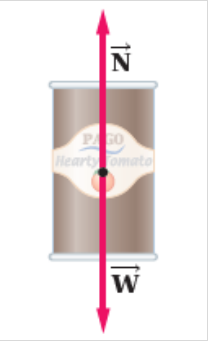
- Consider the action of placing a can of soup on the table.
- Why don't the can of soup accelerate through the table?





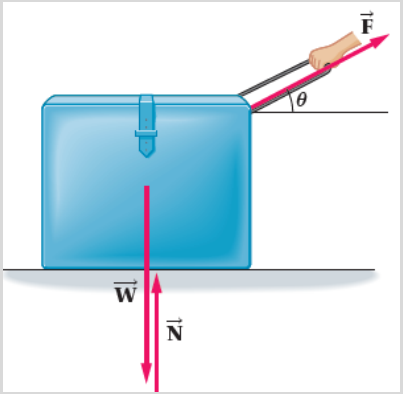
Normal Force

- Since the can is in Static equilibrium, there must be a force exerted on the can by the table to counter the weight of the can.
- This force is called **the normal force**, $\vec{N}$.
- By definition, the direction of the normal force is always perpendicular to the surface where it originates.



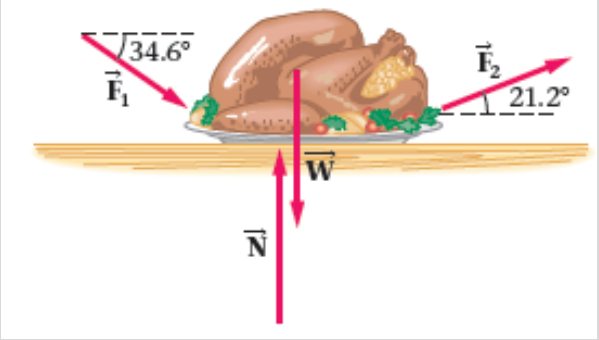
Normal Force

- **The magnitude** of the force is not always equal to the weight of the object. It can be less than, or greater than the weight of the object.
- Consider the forces being exerted on a suitcase being pulled along a surface in the figure below. Is the magnitude of $\vec{N}$ less than, equal to, or greater than the weight $\vec{W}$ of the suitcase?



Normal Force – Example

- **Example:** A 4.53 kg turkey is moved toward the center of a smooth table by two people at a party. Person 1 exerts a force $\overrightarrow{F_1}$ with a magnitude of 35.5 N in the direction indicated. Person 2 exerts a force $\overrightarrow{F_2}$ with a magnitude of 39.7 N in the direction indicated. What is the magnitude $|\vec{N}|$ of the normal force exerted by the table on the turkey?



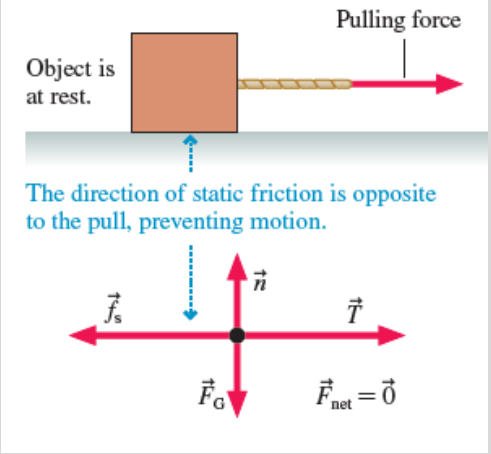
Friction

- **Friction** is an essential force for many things we do such as walking, driving, or sitting down.
- There are **3 main types** of friction: **Static, Kinetic**, and **Rolling**.
- **The direction of friction** is always opposite the **direction of motion**.
- All types of friction are directly proportional to the object's Normal Force.
    - The **Normal Force** of an object the force that the object exerted in the direction normal to the contact surface.

Static Friction

- **Static Friction** is the force acting on an object that keeps it from slipping.
- Static friction is typically calculated with a coefficient of static friction μ_s:

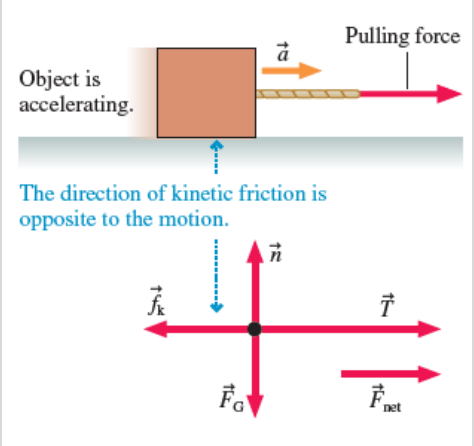
$$f_{s\,max} = \mu_s n$$



Kinetic Friction

- **Kinetic Friction** is the force that replaces Static Friction when the object is already in motion. Kinetic Friction will always be less than Static Friction.
- Similar to Static Friction, Kinetic friction is typically calculated with a coefficient of static friction μ_k:

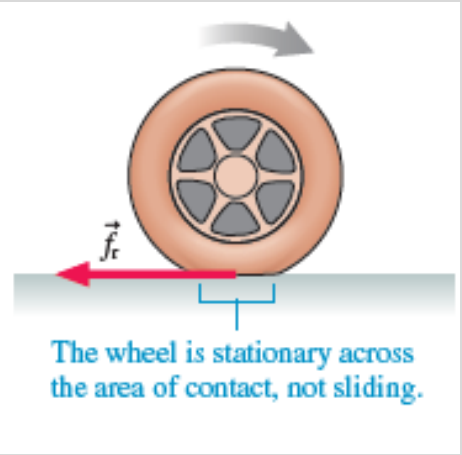
$$f_k = \mu_k n$$



Rolling Friction

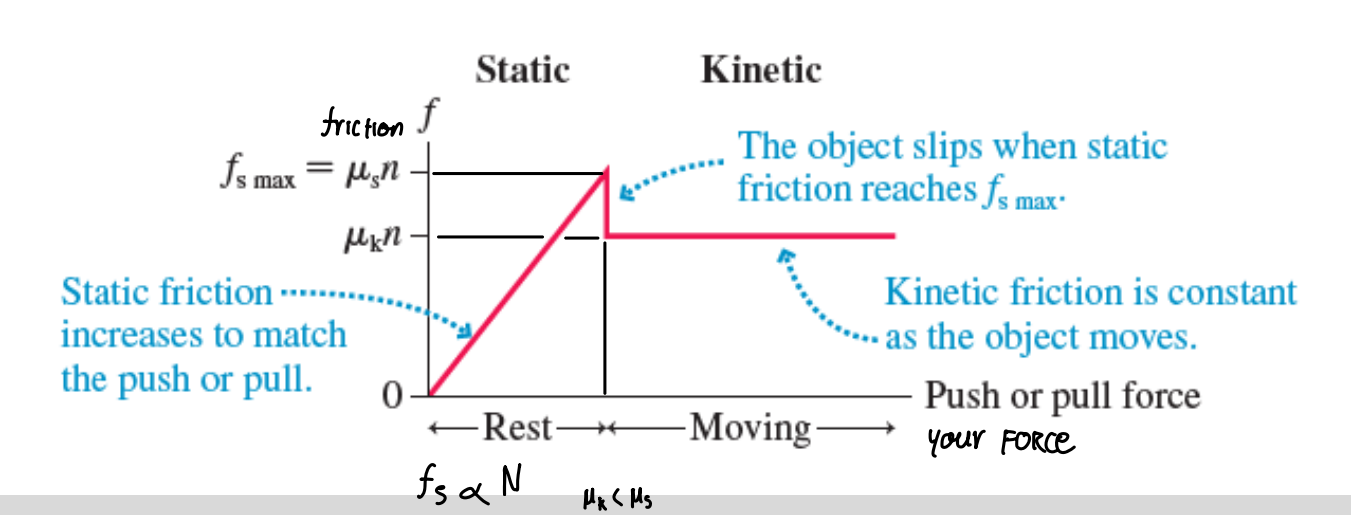
- **Rolling Friction** is the force that is specific to rolling motion. Rolling Friction will always be less than Kinetic Friction.
- Similar to the other 2 types of friction, Rolling friction is typically calculated with a coefficient of Rolling friction μ_r:

$$f_r = \mu_r n$$



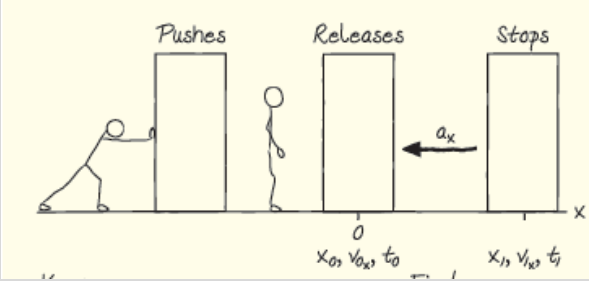
Friction

- **Static friction force** is **zero** if there is no external force being exerted on the object.
- The friction force for an object being pushed can be plotted as:



Friction– Example

- **Example:** Carol pushes a 25 kg wood box across a wood floor at a steady speed of $2.0\ m/s$. How much force does Carol exert on the box? If she stops pushing, how far will the box slide before coming to rest? Given that $\mu_k = 0.2$.



Ropes and Pulley

Tension in a string.

$\mu_k = ?$

$F_{net} = -f_k = m \cdot a_x$

$\uparrow$

$\mu_k N = \mu_k (W)$

$\mu_k = -\frac{m a_x}{mg} = -a_x / g$

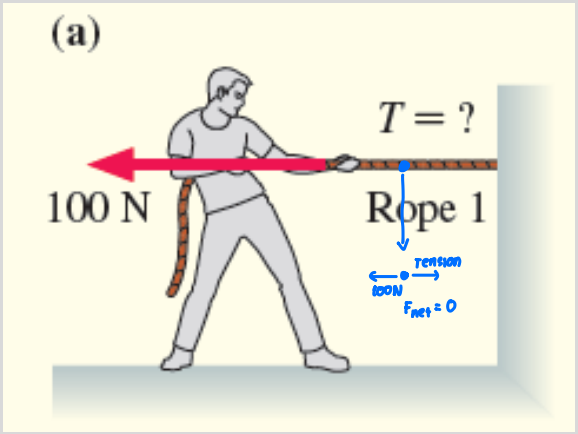
$\mu_k \cdot g = -a_x = \frac{v^2 - v_0^2}{2\Delta x} = \frac{-v_0^2}{2\Delta x}$

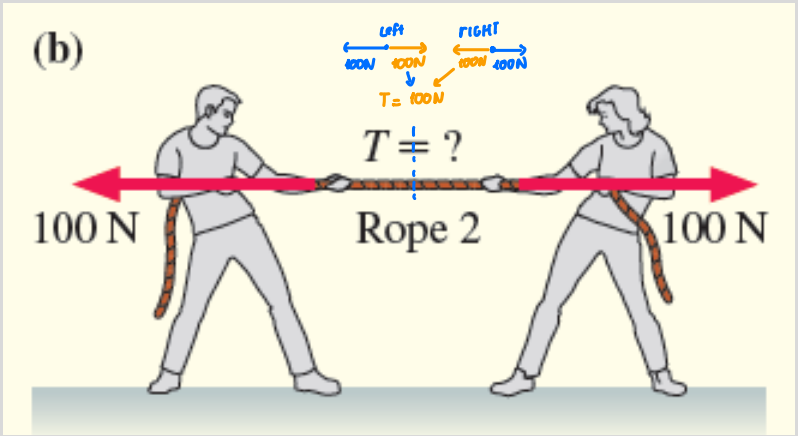
$\mu_k = \frac{v^2}{2\Delta x} \cdot \frac{1}{g} = \frac{v_0^2}{2g\Delta x}$

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Ropes

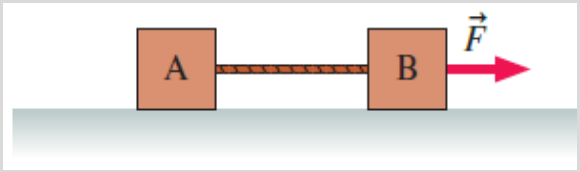
- **Tension** is the pulling force experienced by a rope.
- **Tension** pulls equally in both directions.
  - Example: Examine the following scenarios

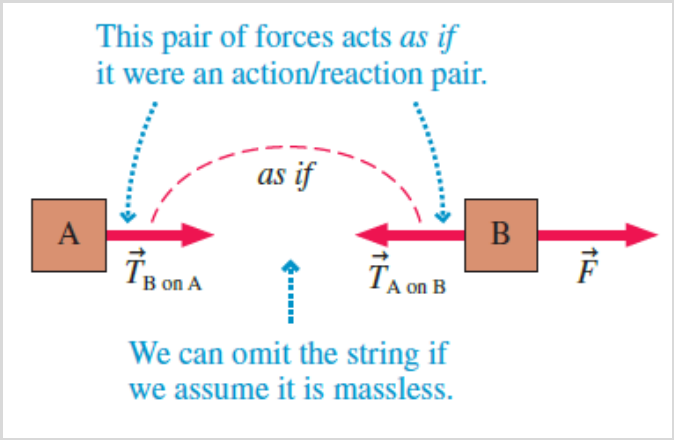




Ropes and Pulleys – Massless String Approximation

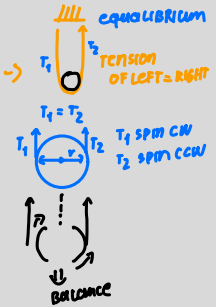
- If objects A and B interact with each other though a massless string, we can omit the string and treat forces $\overrightarrow{F_{A\ on\ B}}$ and $\overrightarrow{F_{B\ on\ A}}$ as if they are an action/reaction pair.





Ropes and Pulleys

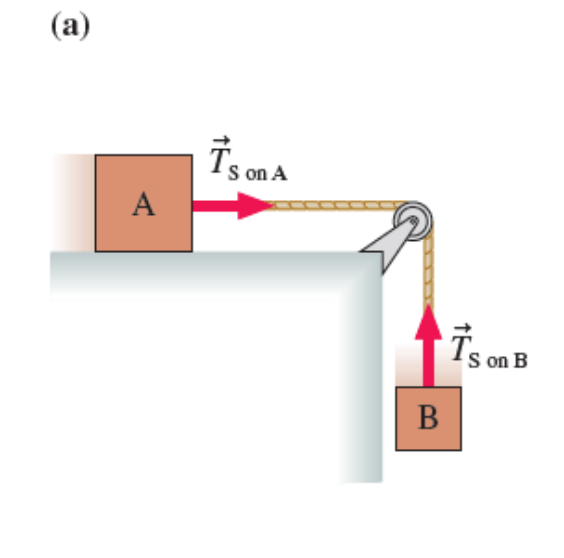
- **Pulleys** are devices that change the direction of a string.
- In **Ropes and Pulleys problems**, we can assume that:
  - The string and the pulley are both <u>massless</u>.
  - There is <u>no friction</u> where the pulley turns on its axle.

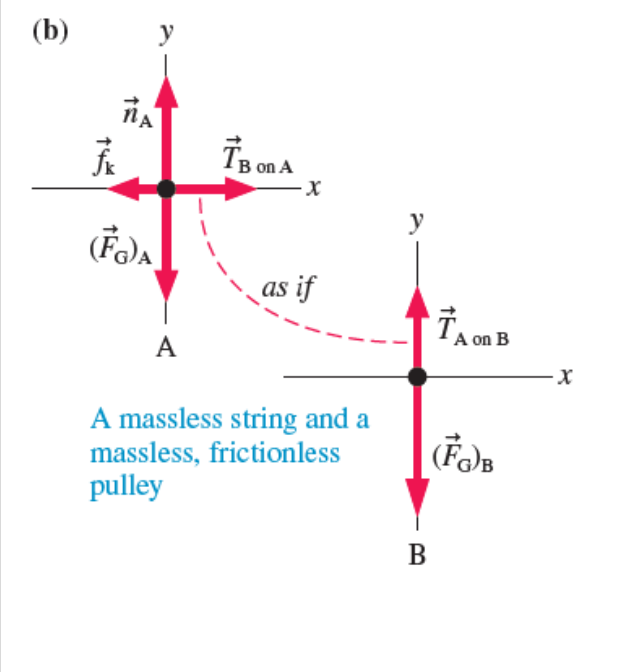






Ropes and Pulleys





Ropes and Pulleys– Example

- **Example:** A 200 kg set used in a play is stored in the loft above the stage (see picture). The director tells a 100 kg employee to lower the set. When he unties the rope, the set falls and the man is pulled into the loft. What is his acceleration?

